# Software Requirements Specification (SRS)

## Internship Application

## 1. Introduction

### 1.1 Purpose

The purpose of this document is to describe the functional and non-functional requirements of **Internship Application**.

This document serves as a guideline for developers, testers, and other stakeholders to ensure that the system meets the needs of students, companies, and school staff during the internship search and application process.

### 1.2 Scope

Internship Application is a web platform designed to manage internship offers and applications.

The system allows students to create profiles, search for internship offers, submit applications, and track their progress. Companies can publish offers and manage received applications, while school staff can monitor students' internship search progress.

### 1.3 Definitions, Acronyms, and Abbreviations

- **IA:** Internship Application
- **CV:** Curriculum Vitae
- **API:** Application Programming Interface
- **RBAC:** Role-Based Access Control
- **GDPR:** General Data Protection Regulation

### 1.4 References

- Business Requirements Document (BRD) – Internship Application
- GDPR requirements for personal data protection

- Course materials and project specifications

### 1.5 Document Overview

This document is structured into several sections: Introduction, General Description, Specific Requirements, Use Case Scenarios, Modeling Diagrams, External Interfaces, and Other Requirements.

Each section provides additional details about the expected behavior and design constraints of Internship Application.

# 2. General Description

### 2.1 Product Perspective

Internship Application is a centralized web application connecting students, companies, and school staff.

The platform replaces part of the internship management currently performed through separate emails, spreadsheets, and external platforms by providing a common system for offers, applications, and student monitoring.

### 2.2 Product Features

Internship Application includes the following main features:

- **Student Profile Management:** Creation and update of student profiles, skills, and CVs.
- **Internship Offer Management:** Creation, modification, publication, and consultation of internship offers.
- **Internship Search:** Search and filtering of available offers.
- **Application Management:** Submission and management of internship applications.
- **Application Tracking:** Monitoring of application statuses.
- **Student Monitoring:** Allows school staff to follow students' internship search progress.
- **Notifications:** Informs users about important application updates.

### 2.3 User Characteristics

The main users include:

- **Students:** Search internship offers, submit applications, and track their progress.
- **Companies / Recruiters:** Publish internship offers and manage received applications.
- **School Staff:** Monitor students and their internship search progress.
- **Administrators:** Manage users, roles, and platform configuration.

### 2.4 General Constraints

- The system must comply with GDPR requirements.
- The system must be accessible through a web browser.
- Access to functions must depend on the user's role.
- Users must have an Internet connection to access the platform.

### 2.5 Assumptions and Dependencies

- Users have basic computer skills.
- Companies provide accurate and up-to-date internship information.
- Students keep their profiles and CVs updated.
- The platform depends on a stable Internet connection.

This follows the same organization as the example SRS, which presents the product perspective, features, user categories, general constraints, and assumptions before detailing the requirements.

# 3. Specific Requirements

## 3.1 Functional Requirements

### 3.1.1 Student Profile Management

- The system must allow students to create and update their profiles.
- Students must be able to add their skills and upload a CV.
- Students must be able to view their application history.

### 3.1.2 Internship Offer Management

- Companies must be able to create, modify, and remove internship offers.
- Internship offers must contain information such as title, description, location, duration, and required skills.
- Students must be able to view available internship offers.

### 3.1.3 Internship Search

- Students must be able to search internship offers.
- The system should allow offers to be filtered by criteria such as location, field, and required skills.

### 3.1.4 Application Management

- Students must be able to apply for internship offers.
- The system must prevent a student from applying multiple times to the same offer.
- Companies must be able to view applications received for their offers.
- Companies must be able to update the status of an application.

### 3.1.5 Application Tracking

- Students must be able to view the current status of their applications.
- Application statuses may include **Submitted, Under Review, Interview, Accepted, and Rejected**.

### 3.1.6 Student Monitoring

- School staff must be able to view students' internship search progress.
- School staff must be able to identify students who have obtained an internship.

### 3.1.7 Notifications

- Students should receive a notification when the status of an application changes.
- Companies should receive a notification when a new application is submitted.

## 3.2 Non-Functional Requirements

### 3.2.1 Performance

- The platform should support multiple simultaneous users.
- Common user actions should be processed within a few seconds under normal conditions.

### 3.2.2 Security

- Communications between the client and server must use HTTPS.
- The system must include role-based access control.
- Users must only access data and features authorized for their role.
- User passwords must be securely stored.

### 3.2.3 Maintainability

- Source code must be documented and organized.
- The application should use a modular architecture.
- Unit and integration tests should be performed during development.

The example SRS similarly separates **functional requirements** from requirements concerning **performance, security, and maintainability**.

# 4. Use Case Scenarios

## 4.1 Use Case 1: Applying for an Internship

**Actors:** Student, System

**Description:**

The student searches for an internship offer and submits an application.

**Steps:**

1. The student logs into the platform.
2. The student searches for internship offers.
3. The system displays available offers.

4. The student selects an internship offer.
5. The student reviews the offer details.
6. The student submits an application.
7. The system records the application.
8. The company is notified of the new application.

**Diagram:**

- **Sequence Diagram:** Illustrates the interaction between the student and Internship Application during the application process.

## 4.2 Use Case 2: Managing an Application

**Actors:** Recruiter, System, Student

**Description:**

The recruiter reviews a student's application and updates its status.

**Steps:**

1. The recruiter logs into the platform.
2. The recruiter accesses the list of received applications.
3. The recruiter selects an application.
4. The system displays the student's information and CV.
5. The recruiter updates the application status.
6. The system saves the new status.
7. The student receives a notification.

**Diagram:**

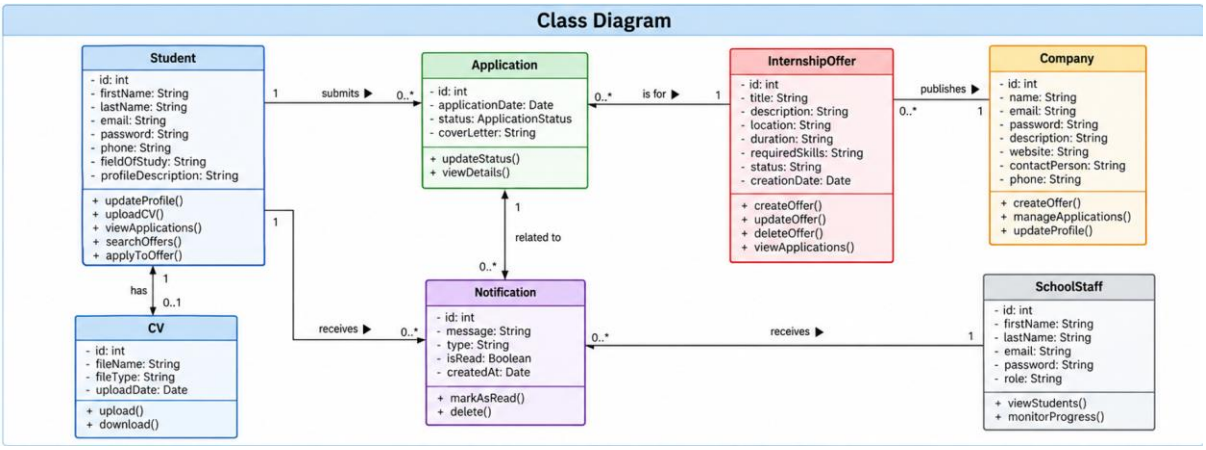
- **Activity Diagram:** Illustrates the recruiter's workflow for reviewing and updating an application.

Comme dans le modèle fourni, deux cas d'utilisation détaillés suffisent à montrer les interactions principales du système.

# 5. Modeling Diagrams

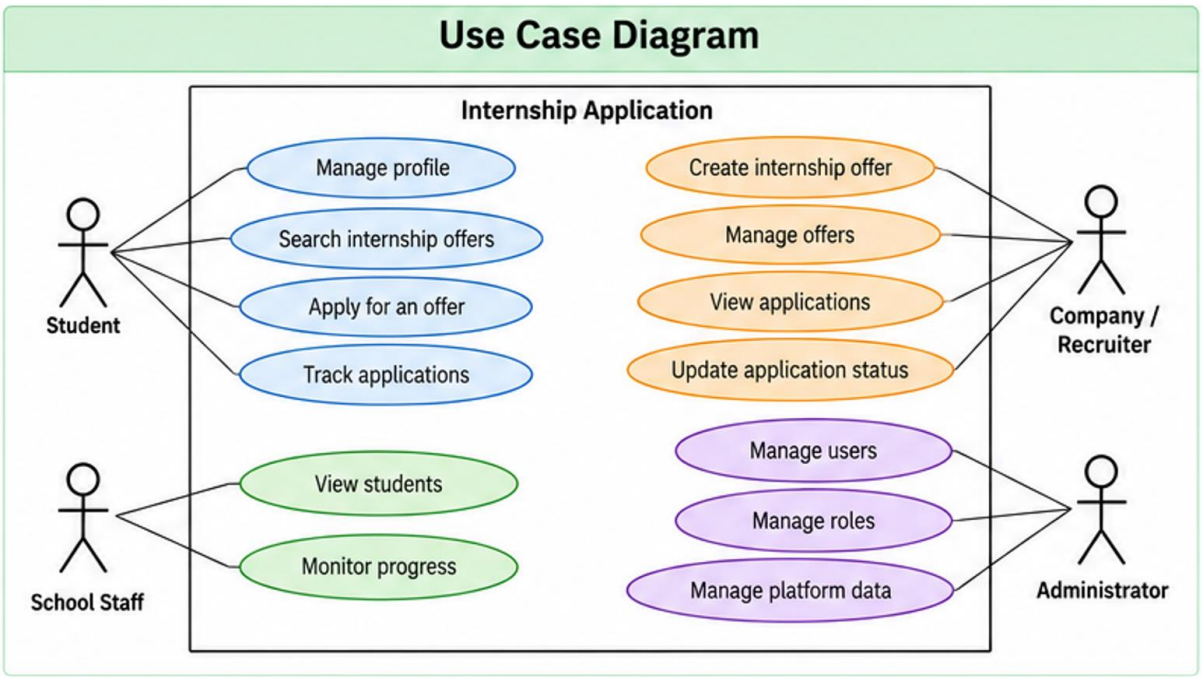
## 5.1 Class Diagram

The following class diagram represents the main entities of Internship Application and their relationships.



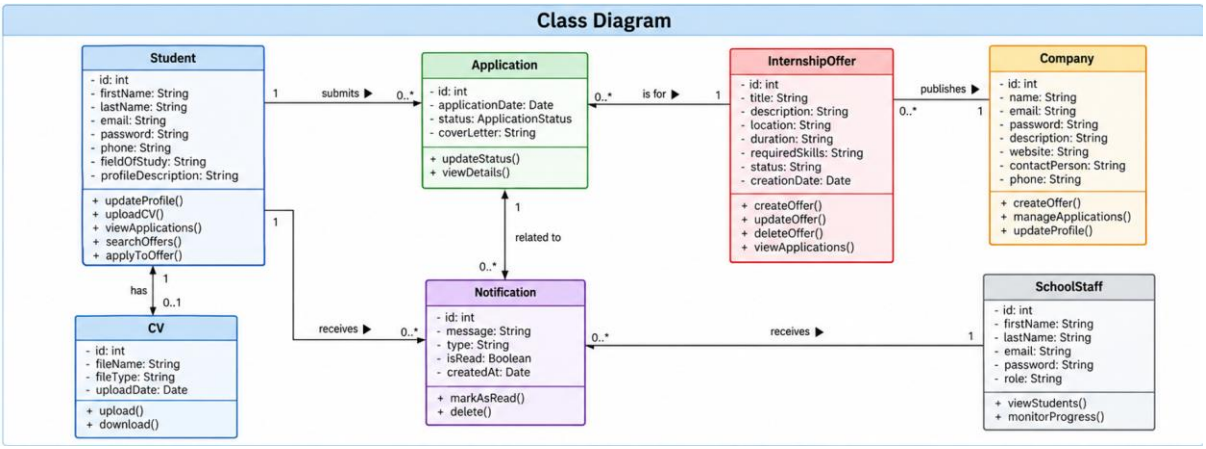

## 5.2 Sequence Diagram

The following sequence diagram illustrates the internship application process.



### 5.3 Use Case Diagram

The following use case diagram represents the main interactions between users and Internship Application.




## 6.1 User Interface

- The system will provide different interfaces depending on the user's role:
- **Student Dashboard:** Allows students to manage their profile, search internship offers, submit applications, and track their status.
- **Recruiter Dashboard:** Allows companies to create internship offers and manage received applications.
- **School Staff Dashboard:** Allows school staff to monitor students' internship search progress.
- **Administrator Interface:** Allows administrators to manage users and roles.

## 6.2 Hardware Interfaces

- No specific hardware interface is required.
- The application should be accessible from standard computers, tablets, and compatible mobile devices.

## 6.3 Software Interfaces

- The frontend will communicate with the backend through RESTful APIs.
- The backend will communicate with a database to store users, offers, and applications.
- The system may use an external email service for notifications.

## 6.4 Communication Interfaces

- The system must use HTTPS for secure communication between clients and the server.
- Data exchanged between the frontend and backend should use JSON format through REST APIs.

The template also separates interfaces into **user, hardware, software, and communication interfaces**.

# 7. Other Requirements

## 7.1 Security and Privacy

- Personal information and uploaded CVs must only be accessible to authorized users.
- The system must comply with GDPR requirements.
- Sensitive information must be protected against unauthorized access.
- Important user actions should be logged when necessary.

## 7.2 Compatibility

- The application should be compatible with modern versions of Google Chrome, Firefox, Safari, and Microsoft Edge.
- The user interface should adapt to different screen sizes.

## 7.3 Maintainability

- The system should use a modular architecture to facilitate future changes.
- New features should be added without significantly affecting existing functionality.
- Source code should be documented and maintained using version control.

These sections correspond directly to the template's **Security and Privacy, Compatibility, and Maintainability** requirements.

# 8. Appendices

- **Appendix A:** Main UML diagrams.
- **Appendix B:** API documentation, if required during development.

# 9. Index

A glossary of the main terms used in the document:

- **Application:** A request submitted by a student for an internship offer.
- **Internship Offer:** An internship opportunity published by a company.
- **Recruiter:** A company representative responsible for publishing offers and managing candidates.
- **Student:** A user searching and applying for internships.
- **School Staff:** Members of the educational institution responsible for monitoring students.
- **Application Status:** Current stage of an application, such as Submitted, Under Review, Interview, Accepted, or Rejected.